

Continuity

ANSWER KEY



Testing continuity

- 1 Consider the piecewise function $f(x) = x^2 + 1$ for $x < 2$, and $f(x) = 3x - 1$ for $x \geq 2$.
- a Find $\lim_{x \rightarrow 2^-} f(x)$. = 5.
- b Find $\lim_{x \rightarrow 2^+} f(x)$. = 5.
- c Is f continuous at $x = 2$? Justify. Yes — both one-sided limits equal $5 = f(2)$, so f is continuous at $x = 2$.
- 2 The piecewise function $f(x) = kx + 1$ for $x \leq 3$, and $f(x) = x^2 - 2$ for $x > 3$, is continuous at $x = 3$. Find the value of k .
- Need $3k + 1 = 9 - 2 = 7$, so $k = 2$.
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Types of discontinuity and the Intermediate Value Theorem

- 3 Classify the discontinuity of each function at the given point:
- a $f(x) = \frac{x^2 - 1}{x - 1}$ at $x = 1$ Removable (a hole) — the factor $(x - 1)$ cancels.
- b $f(x) = \frac{1}{x - 2}$ at $x = 2$ Infinite discontinuity — f blows up as $x \rightarrow 2$.
- c $f(x) = \lfloor x \rfloor$ at $x = 3$ Jump discontinuity — the floor function jumps from 2 to 3.
- 4 Show that $f(x) = x^3 - 4x + 1$ has a root in the interval $(1, 2)$, using the Intermediate Value Theorem.
- $f(1) = 1 - 4 + 1 = -2 < 0$ and $f(2) = 8 - 8 + 1 = 1 > 0$. Since f is continuous (a polynomial) and changes sign, the IVT guarantees a root in $(1, 2)$.
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Continuity on an interval

- 5 State the largest interval(s) on which each function is continuous:
- a $f(x) = \frac{1}{x - 4}$ Continuous everywhere except $x = 4$: on $(-\infty, 4) \cup (4, \infty)$.
- b $f(x) = \sqrt{x - 3}$ Continuous on $[3, \infty)$.
- c $f(x) = \ln(x + 2)$ Continuous on $(-2, \infty)$.
- 6 Find the value of c that makes $f(x) = \frac{x^2 - 9}{x - 3}$ for $x \neq 3$, and $f(3) = c$, continuous at $x = 3$.
- $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \lim_{x \rightarrow 3} (x + 3) = 6$, so $c = 6$.
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Applying the Intermediate Value Theorem

- 7 Show that $f(x) = \cos x - x$ has a root in the interval $(0, 1)$.
 $f(0) = \cos 0 - 0 = 1 > 0$ and $f(1) = \cos 1 - 1 \approx 0.540 - 1 = -0.460 < 0$. Since f is continuous and changes sign on $[0, 1]$, IVT guarantees a root in $(0, 1)$.
- 8 A hiker starts a trail at elevation 200 m at 8:00 AM and finishes at elevation 800 m at noon. Explain, using the IVT, why the hiker must have been at elevation exactly 500 m at some point during the hike.
Elevation as a function of time is continuous (the hiker moves without teleporting), starts below 500 m and ends above 500 m, so by the IVT the hiker's elevation must equal 500 m at some intermediate time.